

Front Powertrain HROE Guarding

Bucket (2024) - Tomasz Krzeminski '26

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Introduction

The front powertrain guarding was based around the 2023 front powertrain guard. The previous design was already a good design, and the general weight to performance of the design was already optimal. The 2022 design was based on an old standard which required a more robust and overdefined system for the guarding. For these reasons, the best source of reference for this part are the 2023 version and the 2024 version due to the large similarity.

The importance of this part comes in the form of user protection as it protects the driver and “maintenance personnel” from injury. Because the powertrain trains are composed of fast acting rotational components, such as the drive train shaft and U-joints, the risk for injury can be severe. The general BAJA handbook refers to the Front and Rear powertrain guards as HROE components, which stands for “Hazardous Release of Energy”.

The last thing of importance to address is general skills and knowledge. The front powertrain guarding requires little to no skill to get started. Everything that is required to design and manufacture this component will be taught throughout the course of this club. The first step of the process is to design the part in Solidworks. I would argue that this is the most important and most time consuming stage as you will spend a lot of time modifying the parts to meet clearances with respect to the other sections of the car. It is **CRITICAL** that you communicate with your Powertrain Lead, as well as the Chassis and Suspension and Ergonomics Leads to avoid improper fitment. This was a major issue during the final assembly of the car.

The following chart is a breakdown of the scoring requirements that must be met to be allowed to clear competition requirements. Everything else that needs to be known about this part can be found in the Baja SAE Handbook.

Specs	5%
Research/ Development	5%
Design / CAD	10%
Data Collection for Analysis	15%
Analysis / Calculations / CAE	20%
Testing / Validation	20%
Correlation of Analysis and Testing	20%
Other	5%

Figure 1: The Design Judge Score Breakdown.

Part History

The powertrain guarding that was referenced the most, and really was the only acceptable design given the requirements was last year's front powertrain guarding. The powertrain guarding prior to last year utilized a design that completely encapsulated the drivetrain and the two U-joints in a steel shell. This was extremely heavy and generally really difficult to manufacture given the amount of welding that was needed to join the steel. It was also difficult to perform general maintenance as the steel shell was well fastened to the chassis of the car. This, along with a change in HROE regulation prompted the design that was adapted in the 2023 iteration of this component.

The 2023 component was manufactured from 6061-T6 aluminum which reduced the weight of the whole guarding system by over 10 pounds. This weight reduction is important in the general performance of the car as a lighter vehicle is capable of moving faster. As

mentioned above, the Front and Rear powertrain guardings were manufactured from an aluminum pipe, and the prop shaft was covered with an acrylic tube that spanned the length of the shaft. Additional fixtures had to be manufactured and mounted on the chassis to hold the prop shaft in the case of failure.

The 2024 iteration of the front powertrain guard utilized the base design of the 2023 version in terms of the general build. The part was made from the same 6061-T6 aluminum pipe, and had aluminum flanges that allowed it to be joined. The part then had a 3D printed fixture which joined to the guarding and then mounted to the footplate to create a stable fixture. This mounting was slightly redesigned from the 2023 version to be slightly smaller, and utilize less material. Finally, the big design change in the current version implemented plastic rivets for quick disassembly. The previous version used nuts and bolts to fixture the components, which wasted a lot of time when maintenance had to be done. The plastic rivets can be ripped off, and because we bought them in bulk, can be replaced whenever they are broken.

My knowledge only goes back a few years. I know that the greatest changes made to this project were last year when we were able to remove a lot of weight from this component by building only the u join guard with metal. It used to be made entirely of metal, covering both the u joints and the prop shaft with steel. We now use a plastic tube to cover the prop shaft and that saved us at least 5 lbs, I think it was closer to 10. Since then we have continued with this design. The only places that nuts and bolts were used was in the footplate fixture, and in one section of the guarding to create a strong fixture between the two halves.

Part Specs (if Necessary)

- The HROE guard must extend 1 in. on each side from the U-joints
- From [Baja SAE rulebook](#):

B.9.2.1 - HROE Guard Materials

All HROE guards shall be constructed of one or both of the following required materials:

- Steel, at least 1.5 mm (0.06 in.) thick, meeting or exceeding the strength of AISI 1010 steel.
- Aluminum, at least 3.0 mm (0.12 in.) thick, meeting or exceeding the strength of 6061-T6 aluminum.

Research

The only research done for this part was referencing last year's design, and finding plastic rivets that would work for holding the parts together. Additionally, referencing the Solidworks assembly is very useful to obtain the necessary measurements. The rear powertrain guarding designer, Michael Jenz, and I found an amazon link for the plastic rivets that perfectly fit our requirements ([This](#) is the link).

Design/CAD

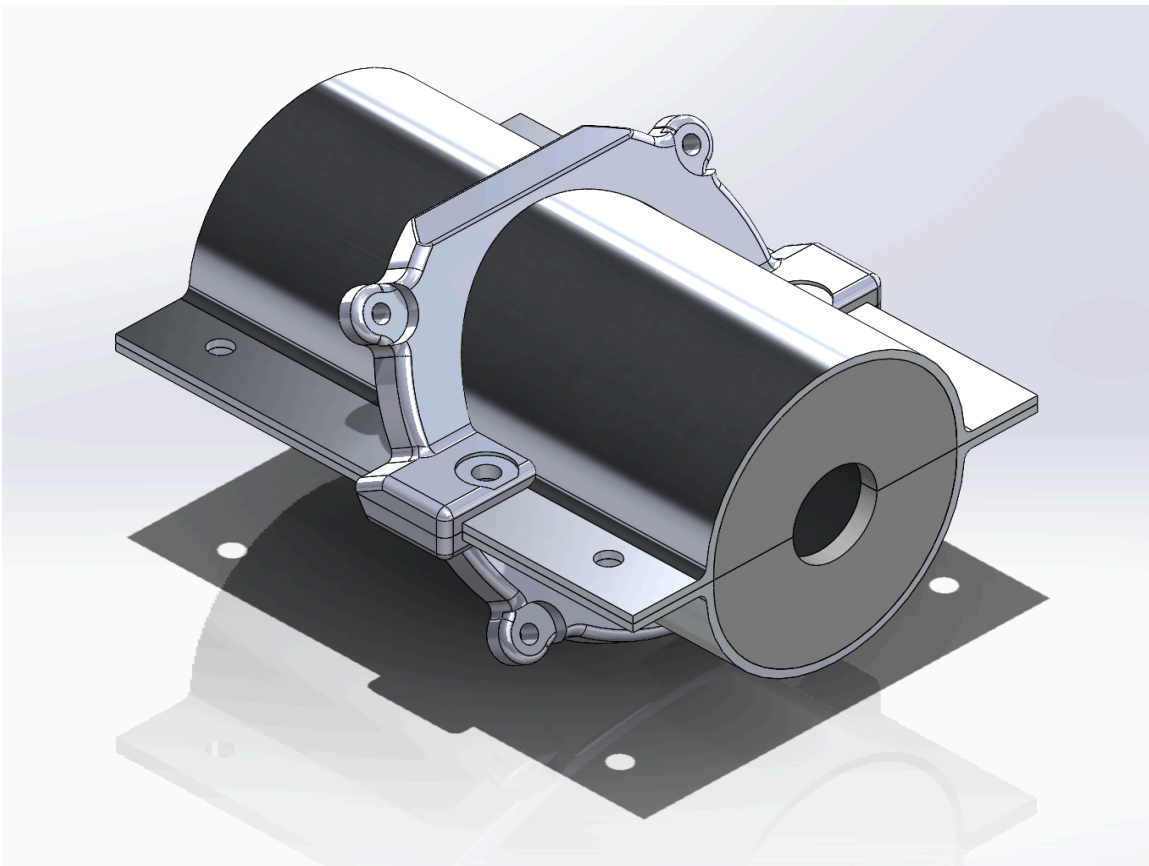


Figure 2: Complete Design

The Front Powertrain Guarding is a 6061-T6 aluminum alloy pipe which has been cut in half and welded to 4 flanges with through holes in them. Figure 2 shows the orthographic view of the complete assembly with the fixtures and the plastic front end. The plastic front end is in place to prevent finger access inside the guarding. The hole in the

middle fits around the acrylic tube, and restricts access from the outside. The holes on the flanges allow for the bolts and plastic rivets to secure the guarding around the U-joint. Finally, the fixture, which is seen at an angle, has 4 holes that firmly secure the guarding to the footplate. This restricts the guard from moving radially around the U-joints, and axially along the U-joint.

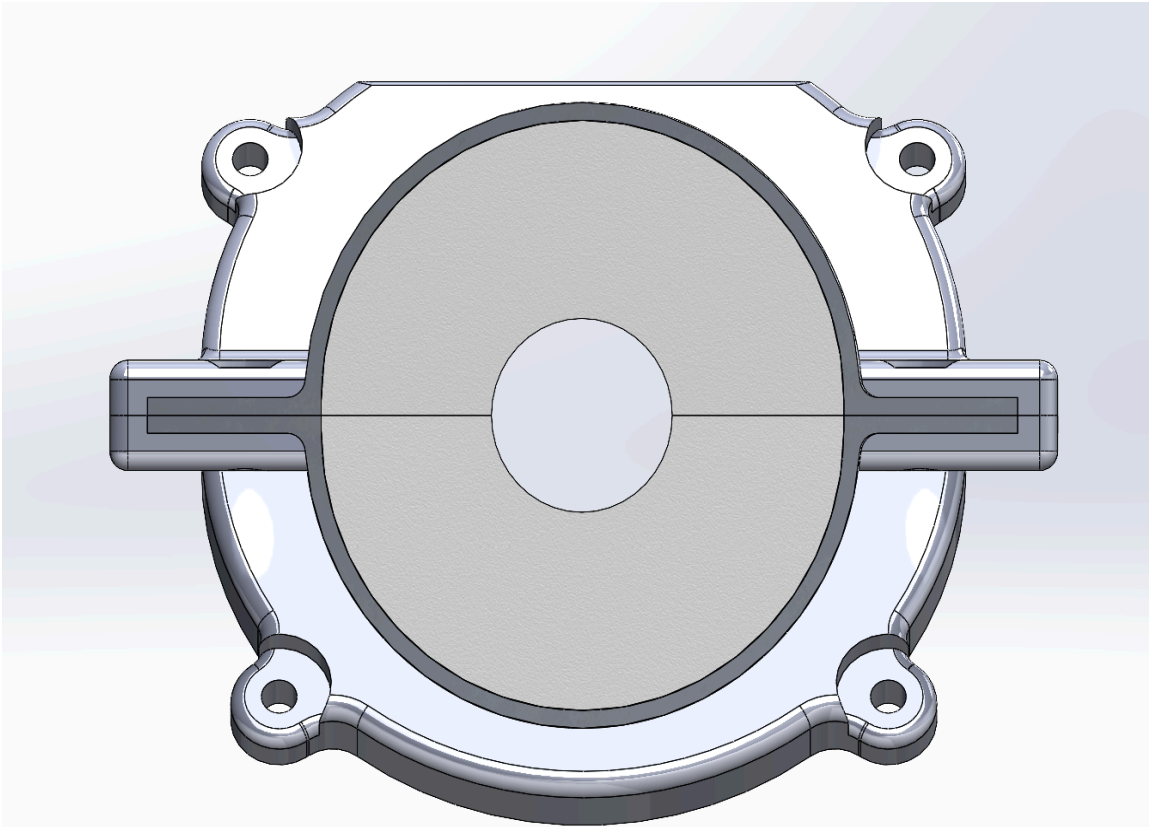


Figure 3: Front View

The front view of the guarding paints pretty much the same picture as the orthographic view, however, it better portrays the general symmetry of the component. Additionally, it is seen that there are filets between the flange and aluminum pipe half. This is used to represent the welds that are necessary in this component. This is important to include in the 3D model of the part as you will need this to create a proper fitment between the fixture and the guard. This view also shows that the guarding is not completely cylindrical as each half is stacked on top of two flanges. This creates a displacement, which changes the profile from being circular to be more of an oval.

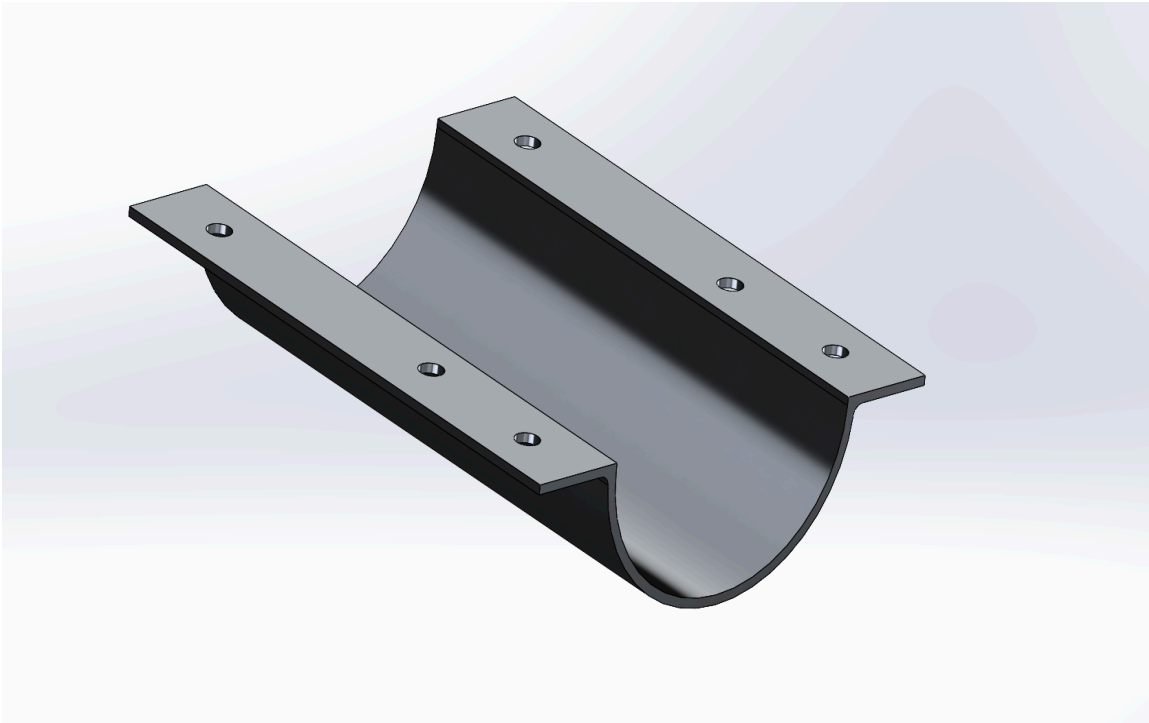


Figure 4: Half of Guard

Figure 4 is used to show the structure of the half pipe and flanges. As can be seen in the figure, the flanges have 3 holes. Each hole corresponds to specific mounting points between the two halves of the guard. These mounting holes were a huge issue as miscommunication between the ergonomics team and powertrain team led to improper mounting. Because of this, the holes that are seen differ from the holes that exist on the physical part.

FEA/Engineering Analysis

No FEA analysis was done on this component.

Manufacturing

1. Cut the aluminum pipe in half using either a hand saw or a bandsaw. The difference between the two methods depends on your approach. If you

create a guide support to prevent the pipe from rotating, you should use the bandsaw. Otherwise, use the hand saw on the pipe, which you should mount on a table vice.

2. Create files necessary to waterjet 4 sets of flanges.
3. Have a welder who knows how to weld aluminum and weld the flanges to the pipe. This step is very important to not mess up, as an incorrect weld will cause you to have to make major corrections to the fitment of the two halves. The mistake that occurred during welding resulted in me having to spend 6-8 hours in the shop to correct and align the two pieces. To avoid having to do the same, overlook the welders when they are assembling your pieces.
4. 3D Print your fixture and front piece.
5. Assemble the completed Front Powertrain Guarding.

Testing/Validation/Performance

Not much testing was done on this part as the only thing it is meant to withstand is a U-joint failure. This in itself is already highly unlikely from happening, so there is very little chance that the guarding will blow up. The only other testing that needs to be done is to test whether your fingers can reach any of the powertrain components. Obviously only perform this testing when the car is off to not lose your fingers.

Correlation of Analysis and Testing

The part worked as intended, however, there were issues with mounting it on the car because of the miscommunication that occurred between subteams. This resulted in the powertrain crew having to bash and grind away material on the front powertrain guarding until it finally fit on the car without hitting any other parts. Additionally, there was a small requirement which this design didn't meet. This requirement is that the end of the guarding had to have a mesh or some sort of fixture that prevented finger access on the end side of the guarding. This was overlooked in this year's design as it was unclear

whether this was needed until tech inspection came around. So don't be like me and design the other end of the guarding.

Mistakes/Takeaways/Future Development

If I was designing this part again, I would ask others how to do Solidworks properly. I did a lot of messing around with Solidworks to get the design to work as I wanted it to. This took a lot of time, and caused a lot of pain when Solidworks decided that it no longer wanted to work. So, to prevent this issue from happening, talk to your subteam members, or watch good tutorials. For manufacturing, I recommend designing processes that can help in the general process of cutting and welding the parts. I have some ideas that could be of use, so if you have any questions just ask me. The way that we cut the parts this year were very inefficient and time consuming.